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DEPARTMENT OF INFORMATION AND COMMUNICATION
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A Project Report on

Free Space Optical Communication System Using Laser Diode and LDR for Morse Code Transmission

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Abstract

This paper presents the design, implementation, and characterization of a low-cost optical fiber communication system for short-range data transmission between two laptops. The system utilizes a 650 nm red laser diode module as the optical source, a Cadmium Sulfide (CdS) Light Dependent Resistor (LDR) as the photodetector, and Arduino Uno R3 microcontrollers for signal processing and interfacing. Initial attempts to establish communication using standard Universal Asynchronous Receiver-Transmitter (UART) serial protocols at baud rates of 300 and 50 were unsuccessful due to the slow switching latency of the LDR, which caused severe inter-symbol interference (ISI). To overcome this physical limitation, an On-Off Keying (OOK) modulation scheme based on International Morse Code was implemented with a base pulse width of 500 ms. The transmitter encodes typed characters into optical pulses, which are coupled into a plastic optical fiber (POF) and decoded at the receiver using a digital comparator-based threshold detection circuit. Experimental results demonstrate successful transmission of alphabetic characters with 100 % accuracy after calibration, validating the proposed approach as an effective educational platform for demonstrating optical communication principles, including source-to-fiber coupling, signal attenuation, and the trade-off between detector bandwidth and data rate.

Index Terms—Optical fiber communication, free-space optics, Arduino, light dependent resistor, Morse code, on-off keying, inter-symbol interference.

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1. Introduction

Optical fiber communication has become the backbone of modern telecommunications, offering high bandwidth, low attenuation, and immunity to electromagnetic interference [1]. While commercial systems employ sophisticated laser diodes, photodetectors, and high-speed modulation schemes, educational implementations often rely on low-cost components to demonstrate fundamental physical-layer concepts [2].

This project investigates the feasibility of establishing a digital communication link between two laptops using a 650 nm continuous-wave laser module, a CdS-based LDR sensor module, and Arduino Uno microcontrollers [3]. The objective is to transmit text data through an optical channel—initially via direct serial communication, and subsequently through a robust, low-speed modulation scheme designed to accommodate the severe bandwidth limitations of the photoresistive detector.

The remainder of this paper is organized as follows: Section 2 describes the system architecture and component specifications. Section 3 details the methodology, including the transition from UART to Morse-code-based On-Off Keying (OOK). Section 4 presents the experimental results and discusses the challenges encountered. Section 5 concludes the work [2].

2. System Architecture and Components

A. Hardware Overview

The proposed link consists of two identical hardware setups, each comprising an Arduino Uno R3, a 650 nm laser module, and an optical sensor module, as summarized in Table I [2][4].

Table I: System Components

SI No.	Component	Specification	Function
1	Microcontroller	ATmega328P (Arduino Uno R3)	Encoding/decoding, USB-UART bridge
2	Optical Source	650 nm, 5V Laser Diode Module	Electrical-to-optical conversion
3	Detector	CdS LDR Sensor Module (LM393)	Optical-to-electrical conversion
4	Transmission Medium	Plastic Optical Fiber (POF)	Guided optical channel
5	Interface	USB Type-B	Laptop-to-Arduino serial communication

The laser module operates at a nominal wavelength of 650 nm in the red visible spectrum. The detector module integrates a CdS photoresistor with an LM393 voltage comparator, providing a digital output (TTL logic) via an adjustable threshold potentiometer.

B. Channel Characteristics

The optical channel is established by butt-coupling the laser output into one end of a plastic optical fiber and aligning the opposing end directly onto the active area of the LDR. Black electrical tape and opaque shielding are employed to suppress ambient light noise at the receiver aperture [1].

3. Methodology

A. Initial Approach

UART Serial Communication The first design iteration attempted to utilize the Arduino's hardware UART (Pins 0 and 1) and SoftwareSerial library to transmit standard ASCII characters at baud rates of 9600, 300, and 50. However, this approach failed due to the rise/fall time limitations of the CdS LDR. The photoresistive element exhibits a temporal response on the order of 20–50 ms, whereas a 50 baud UART bit period is only 20 ms. Consequently, the detector could not resolve individual bits, resulting in inter-symbol interference (ISI) and erroneous character decoding (e.g., the transmission of 'k' yielding '=J=') [3].

B. Proposed Solution

Morse Code OOK Modulation To circumvent the detector bandwidth bottleneck, the system was redesigned to use a symbol-duration-based protocol. International Morse Code was selected because its temporal structure—short marks (dots) and long marks (dashes)—is inherently tolerant of detector latency, provided the base time unit is sufficiently large [4].

The implemented timing parameters are defined as follows:

- Dot (•): Laser ON for $T = 500$ ms
- Dash (–): Laser ON for $3T = 1500$ ms
- Intra-character gap: Laser OFF for $T = 500$ ms
- Inter-character gap: Laser OFF for $3T = 1500$ ms

C. Transmitter Design

The transmitter Arduino monitors its hardware serial port (USB) for incoming ASCII characters from Laptop A. Upon receiving a character, it is converted to uppercase and mapped to its corresponding Morse string. A function `sendMorse()` modulates the laser by driving Digital Pin 13 HIGH/LOW according to the dot-dash timing protocol defined above [2][4].

D. Receiver Design

The receiver Arduino samples Digital Pin 2, which is connected to the LDR module's digital output. A state-machine-based algorithm measures the duration of each light pulse using the `millis()` timer:

- If $50 \text{ ms} < \text{duration} < 700 \text{ ms}$, the pulse is classified as a dot.
- If $\text{duration} \geq 700 \text{ ms}$, the pulse is classified as a dash.

A static timer detects the absence of light for longer than 1500 ms, signaling the end of a character. The accumulated dot-dash string is then passed to a lookup function that returns the decoded ASCII character, which is printed to Laptop B's serial monitor [2][4].

E. Calibration and Signal Conditioning

Prior to data transmission, the LDR module's potentiometer was calibrated such that the onboard comparator LED triggered exclusively when the coupled fiber output exceeded a threshold irradiance. This step was critical to prevent ambient room lighting from creating a false DC bias at the receiver input [2].

4. Results and Discussion

A. Experimental Validation

The system was tested by transmitting individual alphabetic characters from Laptop A to Laptop B via a 1-meter plastic optical fiber. Table II presents a sample of the decoded output [4].

Table II: Sample Transmission Results

Transmitted Character	Morse Pattern	Received Output	Status
A	.-	A	Success
E	.	E	Success
K	-. -	K	Success
L	.- ..	L	Success
S	...	S	Success
T	-	T	Success

Following software refinement of the decoder logic, the system achieved 100% character recognition accuracy under stable alignment conditions.

B. Challenges Encountered

1. **Detector Latency:** The CdS LDR's slow response fundamentally limited the channel capacity. Attempts to push the data rate above approximately 10 bits per second resulted in symbol smearing and decoding errors.
2. **Logic Inversion:** The LDR module's digital output was found to be active-low (logic LOW when illuminated). The receiver firmware was adapted to account for this inversion.
3. **Mechanical Alignment:** The 650 nm laser produces a narrow beam. Misalignment between the fiber face and the laser aperture caused insertion losses exceeding the receiver sensitivity, resulting in missed symbols.
4. **Ambient Light Sensitivity:** Without opaque shielding, fluorescent room lighting introduced a low-frequency noise floor that biased the comparator threshold.

C. Performance Metrics

- **Modulation:** OOK (Morse Code)
- **Base Symbol Rate:** 0.5 s per unit interval
- **Effective Data Rate:** Approximately 2–3 characters per second (variable with letter complexity)
- **Bit Error Rate (BER):** ~0 % after calibration (for static alignment)

5. Conclusion

This project successfully demonstrates a functional, low-cost optical fiber communication link using consumer-grade components. While the CdS LDR proved inadequate for standard UART serial protocols due to its inherent switching latency, the adoption of a Morse-code-based OOK modulation scheme provided a robust and educationally valuable alternative. The system effectively illustrates core concepts of optical communications, including electro-optical conversion, fiber-to-detector coupling, threshold detection, and the critical relationship between detector bandwidth and maximum achievable data rate. Future improvements could include replacing the LDR with a photodiode or phototransistor to enable kilobaud-rate serial transmission, as well as implementing Manchester encoding for DC balance and clock recovery [1] [3].

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